Co-ordination and response

Mark scheme and worked solutions for 20 questions: 7 multiple choice, 13 structured.

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1 · Part 6

Paper 1 & 2 Co-ordination and response [1]

Answer: B

Worked solution

Think about what has to happen first: nothing can respond until something has changed (stimulus) and that change has been detected (receptor). Only then can an effector, such as a muscle or gland, bring about the response.

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2 · Part 6

Paper 1 & 2 Co-ordination and response [1]

Answer: D

Explanation

The central nervous system (CNS) is the brain and the spinal cord. Sense organs and nerves are not part of the CNS: nerves link the sense organs to the CNS, so A, B and C are wrong.

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3 · Part 6

Paper 1 & 2 Co-ordination and response [1]

Answer: A

Worked solution

Follow the signal in time: electrical impulse arrives → chemical released → chemical crosses gap → chemical binds to the next neurone, which starts a new impulse there.

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4 · Part 6

Paper 1 & 2 Co-ordination and response [1]

Answer: C

Worked solution

Near object: ciliary muscles contract → suspensory ligaments slacken → lens becomes fatter. Distant object: the opposite of each step.

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5 · Part 6 Paper 2 only Co-ordination and response [1]

Answer: B

Explanation

ADH is released by the pituitary gland and oestrogen is produced by the ovaries. ADH acts on the kidneys but is not made there (C, D); oestrogen is not made by the testes (A) or the uterus (D), although it acts on the uterus.

6 · Part 6 Paper 1 & 2 Co-ordination and response [1]

Answer: D

Worked solution

Bending needs two things at the tip: auxin to be made there, and light from one side reaching the tip so that auxin collects on the shaded side. Removing the tip removes both; covering the tip stops the light reaching it.

7 · Part 6 Paper 2 only Co-ordination and response [1]

Answer: A

Explanation

LH (from the pituitary gland) stimulates ovulation, the release of an egg from an ovary. FSH causes an egg to mature, oestrogen causes the lining of the uterus to thicken, and progesterone maintains the lining, so B, C and D are wrong.

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8 · Part 6

Paper 1 & 2 | Co-ordination and response | [7]

Mark scheme

(a) **[1]**

- maintenance of a constant internal environment (1)

ALLOW keeping conditions inside the body constant

(b) **[2]**

- body temperature (1)
- body water content (1)

ALLOW blood glucose concentration for one mark

IGNORE "keeping warm" without reference to body temperature

(c) **[2]**

- stimulus: dry air OR low humidity OR dryness of the air (1)
- effector: (leg) muscles (1)

ALLOW legs for effector

REJECT damp air as the stimulus

(d) **[2]**

- the woodlouse moves quickly out of dry places and stays in damp places (1)
- so it loses less water (by evaporation) OR does not dry out (and die) (1)

Worked solution

Homeostasis means keeping the internal environment constant; the specification gives body temperature and body water content as the two examples. In the woodlouse, the change it detects (the stimulus) is dry air. The response is moving faster, which is carried out by its muscles (the effector). Moving fast in dry air and stopping in damp air means it spends most time in damp places, so it loses less water through its body surface and does not dry out.

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9 · Part 6

Paper 1 & 2 | Co-ordination and response | [6]

Mark scheme

(a) **[2]**

- the root grows (curves) downwards (1)
- the shoot grows (curves) upwards (1)

REJECT "grows" with no direction

(b) **[1]**

- geotropic response (1)

ALLOW geotropism OR positive geotropism (root) and negative geotropism (shoot)

ALLOW gravitropism

(c) **[2]**

- light (from one side) would cause a phototropic response OR would affect the direction of growth (1)
- so the only stimulus affecting the direction of growth is gravity (1)

ALLOW so that the results are valid OR it is a fair test, only as the second mark when linked to gravity

(d) **[1]**

- the roots grow away from the light (1)

ALLOW negative phototropism

Worked solution

With the seedling on its side, gravity acts at right angles to the root and shoot. Roots grow towards gravity, so the root curves down; shoots grow away from gravity, so the shoot curves up. This is a geotropic response. Keeping the seedlings in the dark removes light as a second stimulus, so any change in direction must be caused by gravity. When light comes from one side, roots grow away from it.

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10 · Part 6

Paper 1 & 2 | Co-ordination and response | [7]

Mark scheme

(a) **[2]**

- as the number of tips increases, the angle of bending increases (1)
- the increase becomes smaller OR levels off at higher numbers of tips (1)

ALLOW data quoted for the second mark, e.g. increase of only 1 degree from 3 to 4 tips

(b) **[1]**

- so that no auxin came from the shoot's own tip OR so that the only auxin present came from the agar block (1)

(c) **[1]**

- to show that the agar alone does not cause bending OR as a control (1)

ALLOW to show that the bending is caused by a substance from the tips

(d) **[3]**

- auxin (made in the tip) moves to the shaded side of the shoot (1)
- auxin causes the cells on the shaded side to elongate more OR to grow longer (1)
- so the shaded side grows faster than the lit side and the shoot bends towards the light (1)

REJECT auxin moves towards the light

Worked solution

(a) From 0 to 3 tips the angle rises from 0 to 17 degrees, but adding a fourth tip only adds 1 degree, so the effect levels off. (b) A shoot's own tip would release its own auxin, which would change the result. (c) The block with no tips shows that agar on its own does not make the shoot bend. (d) In a shoot lit from one side, auxin collects on the shaded side. Auxin makes cells elongate, so the shaded side grows longer than the lit side and the shoot curves towards the light.

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11 · Part 6 Paper 1 & 2 Co-ordination and response [6]

Mark scheme

(a) **[4]**

- type of signal, hormonal: chemical OR hormones (1)
- how carried, hormonal: in the blood (1)

ALLOW in the plasma

- speed, hormonal: slower (1)
- how long the effect lasts, nervous: short OR brief (1)

(b) **[2]**

- hormones are carried in the blood, which flows to all parts of the body (1)
- any organ that responds to the hormone (a target organ) will be affected (1)

Worked solution

Nervous communication uses electrical impulses carried along neurones; the response is rapid and short-lived. Hormonal communication uses chemicals (hormones) carried in the blood plasma; the response is slower but lasts longer. Because blood reaches every organ, one hormone can reach many organs, and every organ that responds to it is affected.

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12 · Part 6 Paper 1 & 2 Co-ordination and response [6]

Mark scheme

(a) **[4]**

- receptors (in the skin) detect the heat (stimulus) (1)
- an (electrical) impulse passes along a sensory neurone to the spinal cord (1)
- the impulse passes to a relay neurone in the spinal cord and then to a motor neurone (1)
- the motor neurone carries the impulse to a muscle (effector), which contracts and moves the finger away (1)

ALLOW impulses cross synapses between the neurones

ALLOW CNS for spinal cord

(b) **[2]**

- the response is rapid OR automatic (does not need conscious thought) (1)
- so less damage is done to the skin OR the burn is less serious (1)

Worked solution

The heat is the stimulus. Receptors in the skin of the finger detect it and send an electrical impulse along a sensory neurone to the spinal cord. There the impulse passes, across synapses, through a relay neurone to a motor neurone. The motor neurone carries the impulse to a muscle in the arm, the effector, which contracts and pulls the finger away. Because the impulse does not have to wait for a conscious decision, the response is very fast, so the finger is in contact with the hot pan for less time and less damage is done.

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13 · Part 6 Paper 1 & 2 Co-ordination and response [8]

Mark scheme

(a) **[2]**

- independent variable: whether or not Jasmine was listening to music (1)
- dependent variable: reaction time (1)

ALLOW distance the ruler fell for the dependent variable

(b) **[2]**

- any two of:
- same person dropping the ruler (1)
- same ruler (1)
- same hand used to catch the ruler (1)
- same starting position of the ruler (zero mark level with the thumb) (1)
- same volume of music OR same piece of music (1)
- same position of Jasmine's arm, e.g. resting on a table (1)

IGNORE "same conditions" without a variable named

(c) **[2]**

- (0.25 + 0.24 + 0.26 + 0.25) ÷ 4 with 0.38 left out (1)
- 0.25 s (1)

ALLOW 0.28 s (mean of all five results) for one mark

(d) **[2]**

- impulses have to travel to the brain OR a conscious decision is needed (1)
- so the pathway is longer OR has more neurones OR more synapses than a reflex arc (1)

ALLOW a reflex arc passes through the spinal cord without involving conscious thought

Worked solution

(a) Jasmine changed whether music was playing (independent) and measured reaction time (dependent). (b) Anything else that could change reaction time, such as the hand used, the person dropping the ruler or the volume of the music, must stay the same. (c) 0.38 s is much larger than the other results, so it is anomalous. Total of the other four = 0.25 + 0.24 + 0.26 + 0.25 = 1.00 s; mean = 1.00 ÷ 4 = 0.25 s. (d) In this test the eyes detect the falling ruler and impulses travel to the brain, where a decision is made before impulses go to the muscles. This pathway is longer and has more synapses than a reflex arc, so it takes more time.

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14 · Part 6 Paper 1 & 2 Co-ordination and response [6]

Mark scheme

(a) **[4]**

- cornea: refracts (bends) light as it enters the eye (1)

ALLOW transparent layer that lets light in and helps focus it

- iris: controls the amount of light entering the eye OR controls the size of the pupil (1)
- retina: contains light-sensitive (receptor) cells OR detects light (1)

ALLOW rods and cones

- optic nerve: carries (electrical) impulses from the eye (retina) to the brain (1)

(b) **[2]**

- the eye detects a stimulus (light) (1)
- and sends electrical impulses to the brain OR CNS (1)

Worked solution

Light enters through the cornea, which refracts it. The iris changes the size of the pupil to control how much light gets in. The retina at the back of the eye contains light-sensitive receptor cells, and the optic nerve carries impulses from the retina to the brain. A receptor is something that detects a stimulus and starts impulses; the eye detects light and sends impulses to the brain, so it is a receptor.

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15 · Part 6

Paper 1 & 2 | Co-ordination and response | [6]

Mark scheme

(a) **[1]**

- as light intensity increases, the pupil diameter decreases (1)

(b) **[2]**

- 7.0 ÷ 2.4 (1)
- 2.9 (times) (1)

ALLOW 2.92 OR 3 (times) for full marks if working shown

(c) **[2]**

- circular muscles (of the iris) contract (1)
- radial muscles (of the iris) relax (1)

(d) **[1]**

- less light enters the eye, so the retina OR light-sensitive cells are not damaged (1)

Worked solution

(a) The data show pupil diameter falling from 7.0 mm to 2.4 mm as light intensity rises. (b) 7.0 ÷ 2.4 = 2.916..., so the pupil is about 2.9 times wider at 10 lux. (c) In bright light the circular muscles of the iris contract and the radial muscles relax, so the pupil becomes smaller. (d) A smaller pupil lets less light in, which protects the retina from damage.

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16 · Part 6 Paper 1 & 2 Co-ordination and response [7]

Mark scheme

(a) **[2]**

- the rate of sweat production increased (from 0.2 to 3.1 g per min) (1)
- it increased quickly for the first 20 minutes and then only slightly OR levelled off (1)

(b) **[2]**

- water in the sweat evaporates (from the skin) (1)
- this transfers heat energy away from the skin OR cools the skin and the blood (1)

(c) **[3]**

- vasodilation (1)
- the blood vessels (arterioles) near the surface of the skin widen (1)
- more blood flows near the skin surface, so more heat is lost (by radiation) to the surroundings (1)

REJECT blood vessels move closer to the surface of the skin

Worked solution

(a) Sweat production rose steeply from 0.2 to 2.9 g per min in the first 20 minutes, then only rose to 3.1 g per min in the last 10 minutes. (b) Sweat is mostly water. When it evaporates from the skin, it takes heat energy with it, which cools the skin and the blood flowing through it. (c) The skin looks redder because of vasodilation: blood vessels near the skin surface widen so more warm blood flows close to the surface, and more heat is lost to the surroundings. The vessels themselves do not move.

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17 · Part 6 Paper 1 & 2 Co-ordination and response [6]

Mark scheme

(a) **[1]**

- adrenal glands (1)

(b) **[2]**

- (88 − 64) ÷ 64 × 100 (1)
- 37.5% (1)

ALLOW 38% OR 37% for full marks if working shown

(c) **[2]**

- more blood is pumped to the leg muscles each minute (1)
- delivering more oxygen and glucose for respiration (to release more energy) (1)

(d) **[1]**

- any one of: increases breathing rate; increases blood glucose concentration; widens (dilates) the pupils; diverts more blood to the muscles (1)

IGNORE increases heart rate (already given)

Worked solution

(a) Adrenaline is made by the adrenal glands. (b) Increase = 88 − 64 = 24 beats per minute; 24 ÷ 64 × 100 = 37.5%. (c) A faster heart rate pumps more blood to the leg muscles every minute, so they receive more oxygen and glucose for respiration and can release more energy for contraction. (d) Adrenaline prepares the body for action in other ways too, such as raising the breathing rate and the blood glucose concentration.

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18 · Part 6 Paper 1 & 2 Co-ordination and response [7]

Mark scheme

(a) **[1]**

- pancreas (1)

(b) **[2]**

- (7.8 − 5.4) ÷ 60 (1)
- 0.04 mmol per dm^3 per min (1)

ALLOW 2.4 mmol per dm^3 per hour for full marks if working shown

ALLOW ECF from an incorrect subtraction with correct division by 60

(c) **[3]**

- the rise in blood glucose concentration causes the pancreas to release (more) insulin (1)
- insulin causes the liver to convert glucose to glycogen OR causes cells to take up glucose from the blood (1)
- so the blood glucose concentration falls back to about 5.0 mmol per dm^3 (its normal level) (1)

(d) **[1]**

- (with very little insulin) less glucose is converted to glycogen OR removed from the blood (1)

Worked solution

(a) Insulin is made by the pancreas. (b) Change = 7.8 − 5.4 = 2.4 mmol per dm^3 over 90 − 30 = 60 minutes; rate = 2.4 ÷ 60 = 0.04 mmol per dm^3 per min. (c) After the drink, the blood glucose concentration of P rises. The pancreas responds by releasing more insulin. Insulin causes the liver to convert glucose into glycogen, so glucose is removed from the blood and the concentration falls back to its normal level of about 5.0 mmol per dm^3 by 120 minutes. (d) Person Q produces very little insulin, so little glucose is converted to glycogen and the concentration stays high.

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19 · Part 6 Paper 2 only Co-ordination and response [8]

Mark scheme

(a) **[2]**

- a smaller volume of urine was produced (1)
- the urine was more concentrated (1)

IGNORE data quoted without a comparison

(b) **[4]**

- any four of:
- the water content of the blood decreased (because of sweating and not drinking) (1)
- this is detected by the brain (1)
- more ADH is released by the pituitary gland (into the blood) (1)
- ADH makes the collecting ducts (kidney tubules) more permeable to water (1)
- more water is reabsorbed (from the kidney tubules) into the blood (1)
- so a smaller volume of more concentrated urine is produced (1)

ALLOW nephrons OR kidney tubules for collecting ducts

(c) **[2]**

- less ADH would be released (1)
- because the water content of the blood increases OR the blood becomes less concentrated (1)

ALLOW so more water is lost in the urine to return the water content to normal for the second mark

Worked solution

(a) In the afternoon Mei made only 20 cm^3 of urine per hour instead of 75 cm^3, and its concentration was much higher (1050 instead of 280 arbitrary units). (b) Sweating and not drinking lowered the water content of her blood. The brain detected this and the pituitary gland released more ADH. ADH travels in the blood to the kidneys and makes the collecting ducts more permeable to water, so more water is reabsorbed into the blood. Less water is left in the urine, so there is a small volume of concentrated urine. This keeps the body water content constant, which is an example of homeostasis. (c) Drinking a lot of water raises the water content of the blood, so less ADH is released, less water is reabsorbed and a larger volume of dilute urine is made.

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20 · Part 6 Paper 2 only Co-ordination and response [7]

Mark scheme

(a) **[2]**

- causes an egg (follicle) to mature in an ovary (1)
- stimulates the ovaries to produce oestrogen (1)

(b) **[1]**

- pituitary gland (1)

(c) **[2]**

- increases from 2 (arbitrary units) at day 14 to a maximum of 12 at day 20 (1)
- then decreases to 2 by day 28 (1)

ALLOW rises then falls with correct days for one mark if no values given

(d) **[2]**

- progesterone maintains the lining of the uterus (1)
- when it falls, the lining breaks down OR menstruation (bleeding) starts (1)

Worked solution

(a) FSH from the pituitary gland causes an egg to mature inside an ovary and stimulates the ovaries to make oestrogen. (b) FSH is released by the pituitary gland. (c) Progesterone stays low until ovulation, then rises from 2 units at day 14 to a peak of 12 units at day 20, before falling to 2 units by day 28. (d) Progesterone keeps the thickened lining of the uterus in place, ready for an embryo. If no pregnancy occurs, progesterone falls, the lining breaks down and is lost in menstruation, which starts the next cycle.

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